

Introduction

Many real-world systems can be represented as networks that change over time. Examples include a communication network with a scheduled transport system, a social network formed by time-constrained interactions, and information on disease-spreading processes. In these cases, time is an important factor in determining how influence and information propagate through the network.

Temporal networks extend graph models by associating time with the edges, supporting the study of time-dependent connectivity. The core concept is the temporal path constraint, which is a sequence of edges in strictly increasing time order. Temporal paths form the basis of temporal connectivity, which determines when a node can be reached from a given source under temporal constraints. This concept is important for understanding mechanisms, like information diffusion, epidemic spreading, and network resistance.

The algorithmic foundations of temporal connectivity are well-studied, including simple and efficient approaches such as temporal breadth-first search (BFS). The existing graph visualisation tools, designed for static networks, provide limited support for exploring the temporal constraints and observing how connectivity evolves over time. This results in a mismatch of temporal networks and tools that are available to explore them.

The Temporal Path Explorer and Visualizer project aims to handle this gap with a web-based tool to explore temporal paths with changing networks. The system will let users upload time-based network data and observe how connectivity from a selected node source evolves over time. Linking the temporal connectivity algorithm with visualisation, this tool makes the concept of time-constrained paths explicit and concrete, enabling users to observe the impact of temporal ordering on network connectivity.

The core of the system will be a simple temporal connectivity algorithm, which will compute reachable nodes stepwise, such as a one-pass temporal BFS. The results are reflected immediately in the visual interface, facilitating users to follow the temporal expansion of reachable nodes. This merging of computation and visualisation supports exploratory analysis and helps to connect the theoretical and practical understanding.

The distinctive feature is that users will be able to remove or delay certain temporal edges and observe how these changes will affect the network. This function of the network will draw attention towards the temporal processes of the events and enable users to experiment with problems, such as delayed communication or missing connections.

The project is primarily implementation-driven and follows an interactive software system. It also includes research on time-varying networks, temporal path algorithms, data structures, and the visual encoding used in the system. The project follows the most recent software development practices, with implementation guided by research and work.

This includes clear requirements analysis, careful system design, and the adoption of an appropriate software development life cycle. The project is carried out by a team, reflecting professional practice in which effective communication, task coordination, and documentation.